# Neural Habitat Modeling: Predicting Species Distribution Under Climate Change

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**Abstract**

Predicting how species distributions shift under climate change is among the most consequential challenges in conservation biology. Classical species distribution models (SDMs) rely on correlative relationships between occurrence records and bioclimatic variables, yet they often fail to capture complex, non-linear habitat dependencies and spatial autocorrelation. We present **HabitatNet**, a graph neural network framework for species distribution modeling that jointly models habitat suitability, landscape connectivity, and migration corridors under projected climate scenarios. HabitatNet operates on a spatial graph representation where nodes encode local environmental features and edges capture dispersal potential. We integrate Shared Socioeconomic Pathway (SSP) climate projections from CMIP6 to forecast range shifts through 2100. Evaluated on 847 terrestrial vertebrate species across six biogeographic realms, HabitatNet achieves a mean AUC of 0.943 and outperforms MaxEnt, random forests, and standard neural SDMs by 8–14% on held-out temporal validation sets. The framework identifies critical connectivity bottlenecks and proposes optimal corridor placements, directly informing conservation prioritization. All models, data pipelines, and results are released under open licenses through the Zoo Open Science Network.

**Keywords:** species distribution modeling, graph neural networks, climate change, habitat connectivity, migration corridors, conservation planning

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# 1 Introduction

Climate change is reshaping the geographic ranges of species at unprecedented rates [Chen et al., 2011, Pecl et al., 2017]. As temperature and precipitation regimes shift, species must either adapt in situ, migrate to newly suitable habitats, or face local extinction [Urban, 2015]. Understanding and predicting these distributional changes is essential for effective conservation planning, protected area design, and biodiversity policy.

Species distribution models (SDMs) have become the primary computational tool for projecting range shifts [Elith & Leathwick, 2009, Guisan & Thuiller, 2005]. Traditional approaches, particularly MaxEnt [Phillips et al., 2006] and generalized additive models [Hastie & Tibshirani, 1990], establish correlative relationships between species occurrence records and environmental covariates. While widely adopted, these models suffer from several well-documented limitations:

(i) **Linearity assumptions**: Most correlative SDMs struggle to capture the complex, non-linear interactions between environmental variables that determine habitat suitability [Elith et al., 2006].

(ii) **Spatial independence**: Standard SDMs treat occurrence records as independent samples, ignoring spatial autocorrelation and dispersal constraints [Dormann et al., 2007].

(iii) **Static landscapes**: Projections typically assume instantaneous equilibrium with climate, ignoring transient dynamics, dispersal limitation, and habitat fragmentation [Zurell et al., 2016].

(iv) **Connectivity neglect**: The physical connectivity of landscapes—whether corridors exist for species to track shifting climate envelopes—is rarely incorporated into distributional predictions [Hodgson et al., 2009].

Recent advances in deep learning offer pathways to address these shortcomings. Neural network-based SDMs can model arbitrary non-linear relationships [Botella et al., 2018, Chen et al., 2020], and graph neural networks (GNNs) provide a natural framework for spatially structured ecological data [Jetz et al., 2019].

## 1.1 Contributions

This paper introduces **HabitatNet**, a graph neural network framework for species distribution modeling that addresses the above limitations. Our contributions are:

1. A **spatial graph representation** of landscapes where nodes encode local environmental features at 1 km resolution and edges represent dispersal potential weighted by terrain resistance.

2. A **multi-species GNN architecture** that shares learned environmental representations across taxa while learning species-specific habitat preferences through attention-weighted readout layers.

3. Integration with **CMIP6 climate projections** under multiple SSP scenarios (SSP1-2.6, SSP2-4.5, SSP3-7.0, SSP5-8.5) for temporal forecasting through 2100.

4. A **corridor identification algorithm** built on learned graph embeddings that identifies critical connectivity bottlenecks and proposes optimal corridor placements.

5. Comprehensive evaluation on **847 terrestrial vertebrate species** across six biogeographic realms, with temporal validation against independent resurvey data.

### 1.2 Relation to Zoo Labs Foundation

Zoo Labs Foundation operates as an open AI research network dedicated to decentralized science (DeSci) and decentralized AI (DeAI). This work advances the Foundation's mission by: (a) developing open-source AI tools for biodiversity conservation, (b) establishing reproducible benchmarks through the Zoo Open Science Network, and (c) enabling community-driven conservation planning through transparent, verifiable models. All artifacts are governed through Zoo Improvement Proposals (ZIPs) at `zips.zoo.ngo`.

## 2 Background and Related Work

### 2.1 Classical Species Distribution Models

The dominant paradigm in SDM is the correlative approach, which relates species presence (and sometimes absence) records to environmental predictors. MaxEnt [Phillips et al., 2006] maximizes entropy of the predicted distribution subject to constraints from observed feature means. Boosted regression trees [Elith et al., 2008] and random forests [Cutler et al., 2007] offer non-parametric alternatives that accommodate interactions among predictors. Ensemble approaches such as BIOMOD2 [Thuiller et al., 2009] combine multiple algorithms to reduce model uncertainty.

Despite their utility, correlative SDMs rest on the assumption that current species-environment relationships are stable under future conditions—the *stationarity assumption*—which climate change directly violates [Dormann et al., 2012].

### 2.2 Deep Learning for SDMs

Neural networks were first applied to SDMs by Lek & Guégan [1999], but modern deep learning architectures have dramatically expanded capability. Convolutional neural networks (CNNs) applied to remotely-sensed imagery can predict species occurrence from raw satellite data [Deneu et al., 2021]. Deep species distribution models (DeepSDM) use multi-layer perceptrons with environmental covariates as inputs [Botella et al., 2018]. However, these approaches treat each spatial location independently and do not leverage landscape structure.

### 2.3 Graph Neural Networks

GNNs generalize deep learning to graph-structured data [Kipf & Welling, 2017, Veličković et al., 2018]. Message-passing neural networks [Gilmer et al., 2017] iteratively update node representations by aggregating information from neighbors. Spatial applications of GNNs include traffic forecasting [Li et al., 2018], molecular property prediction [Duvenaud et al., 2015], and social network analysis [Hamilton et al., 2017]. In ecology, GNNs remain underexplored, though recent work has applied them to food web modeling [Strydom et al., 2021] and metapopulation dynamics [**?**].

### 2.4 Connectivity and Corridor Modeling

Circuit theory [McRae, 2006, McRae et al., 2008] models landscape connectivity as electrical resistance, where current flow between nodes indicates movement probability. Least-cost path analysis [Adriaensen et al., 2003] identifies single optimal routes. Omniscape [Landau et al., 2021] extends circuit theory to continuous surfaces. These approaches require pre-specified resistance surfaces, which are typically hand-crafted from land cover maps—a limitation HabitatNet addresses by learning resistance from data.

# 3 Problem Formulation

## 3.1 Notation

Let $\mathcal{S} = \{s_1, \ldots, s_K\}$ denote a set of $K$ focal species. The study area is discretized into a regular grid of $N$ cells, each represented as a node $v_i$ in a spatial graph $G = (V, E)$, where $V = \{v_1, \ldots, v_N\}$ and $E \subseteq V \times V$.

Each node $v_i$ is associated with an environmental feature vector $\mathbf{x}_i \in \mathbb{R}^d$ comprising bioclimatic variables, topographic metrics, land cover, and vegetation indices. For a climate scenario $c \in \mathcal{C}$ and time period $t$, the features become $\mathbf{x}_i^{(c,t)}$.

Edge weights $w_{ij}$ encode the dispersal permeability between adjacent cells, determined by terrain, land cover, and geographic distance.

## 3.2 Objectives

For each species $s_k$, we seek to learn a function:

$$f_k : (G, \mathbf{X}^{(c,t)}) \to [0, 1]^N \tag{1}$$

that maps the spatial graph with time- and scenario-specific features to a vector of habitat suitability scores across all cells. The model must:

1. Accurately predict current species distributions when validated against independent occurrence data.
2. Produce calibrated probability estimates under future climate projections.
3. Identify spatially connected regions of suitable habitat (connectivity analysis).
4. Locate optimal corridors linking disjoint habitat patches.

## 3.3 Loss Function

Given presence-absence labels $y_i^{(k)} \in \{0, 1\}$ for species $k$ at cell $i$, we minimize a composite loss:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \lambda_1 \mathcal{L}_{\text{spatial}} + \lambda_2 \mathcal{L}_{\text{connect}} \tag{2}$$

where $\mathcal{L}_{\text{BCE}}$ is the weighted binary cross-entropy, $\mathcal{L}_{\text{spatial}}$ penalizes spatially incoherent predictions (encouraging smooth habitat boundaries), and $\mathcal{L}_{\text{connect}}$ rewards predictions that maintain graph connectivity of suitable patches.

# 4 HabitatNet Architecture

## 4.1 Spatial Graph Construction

**Grid resolution.** We discretize the study area into 1 km hexagonal cells, following Jetz et al. [2019]. Hexagonal grids minimize edge effects and provide uniform adjacency [Birch et al., 2007].

**Node features.** Each cell's feature vector $\mathbf{x}_i \in \mathbb{R}^{42}$ comprises:

- 19 bioclimatic variables from WorldClim v2.1 [Fick & Hijmans, 2017]
- 5 topographic variables: elevation, slope, aspect, topographic roughness index (TRI), topographic wetness index (TWI)
- 10 land cover class fractions from ESA WorldCover [Zanaga et al., 2022]

- 4 vegetation indices from MODIS: NDVI mean, NDVI seasonality, LAI, FPAR
- 4 hydrological variables: distance to water, flow accumulation, precipitation seasonality, aridity index

**Edge construction.** Edges connect each cell to its six hexagonal neighbors. Long-range edges (up to 50 km) are added for cells along waterways and ridgelines, reflecting known dispersal corridors.

**Edge weights.** Dispersal permeability $w_{ij}$ is initialized from land cover resistance values but is subsequently refined through learning (Section 4.3).

## 4.2 Network Architecture

HabitatNet consists of four modules:

1. **Feature Encoder**: A shared 3-layer MLP that projects raw environmental features into a latent space:
$$\mathbf{h}_i^{(0)} = \mathrm{MLP}_{\mathrm{enc}}(\mathbf{x}_i) \in \mathbb{R}^{128} \tag{3}$$
2. **Graph Message Passing**: $L$ layers of graph attention [Veličković et al., 2018] with edge-conditioned convolution:
$$\mathbf{h}_i^{(\ell+1)} = \sigma\left(\sum_{j\in\mathcal{N}(i)} \alpha_{ij}^{(\ell)}\, \mathbf{W}^{(\ell)}\mathbf{h}_j^{(\ell)} + \mathbf{W}_e^{(\ell)}\mathbf{e}_{ij}\right) \tag{4}$$
where $\alpha_{ij}^{(\ell)}$ are learned attention coefficients, $\mathbf{e}_{ij}$ is the edge feature vector, and $\sigma$ is the GELU activation.
3. **Species-Specific Readout**: For each species $k$, an attention-weighted readout layer computes habitat suitability:
$$p_i^{(k)} = \sigma_{\mathrm{sig}}\left(\mathbf{w}_k^\top \mathbf{h}_i^{(L)} + b_k\right) \tag{5}$$
where $\mathbf{w}_k \in \mathbb{R}^{128}$ and $b_k \in \mathbb{R}$ are species-specific parameters.
4. **Connectivity Module**: A spectral graph convolution layer that computes Fiedler values [Fiedler, 1973] of the predicted habitat subgraph, enabling differentiable connectivity optimization.

## 4.3 Edge Weight Learning

Rather than relying on hand-crafted resistance surfaces, HabitatNet learns edge weights through a parameterized function:
$$w_{ij} = \sigma_{\mathrm{sig}}(\mathrm{MLP}_{\mathrm{edge}}([\mathbf{x}_i \| \mathbf{x}_j \| \mathbf{g}_{ij}])) \tag{6}$$
where $\mathbf{g}_{ij}$ encodes geographic distance and elevation difference, and $\|$ denotes concatenation. This allows the model to discover species-group-specific resistance patterns from occurrence data.

### 4.4 Climate Projection Integration

For future projections, we replace the 19 bioclimatic variables with downscaled CMIP6 projections [Eyring et al., 2016]. We use the ensemble median of 12 general circulation models (GCMs) under four SSP scenarios. The remaining features (topography, land cover) are kept static for the base analysis but can be updated with projected land-use change scenarios from LUH2 [Hurtt et al., 2020].

To account for GCM uncertainty, we propagate the full ensemble through HabitatNet and report the median and interquartile range of suitability predictions.

## 5 Corridor Identification Algorithm

Given the predicted habitat suitability map $\mathbf{p}^{(k)} \in [0,1]^N$ and learned edge weights $\{w_{ij}\}$, we identify migration corridors through the following procedure.

### 5.1 Habitat Patch Detection

We threshold the suitability map at $\tau = 0.5$ and extract connected components as habitat patches $\{P_1, \dots, P_M\}$. Patches smaller than a species-specific minimum area requirement are removed.

### 5.2 Inter-Patch Connectivity

For each pair of patches $(P_a, P_b)$, we compute the effective resistance $R_{ab}$ using the learned edge weights as conductances:

$$R_{ab} = (\mathbf{e}_a - \mathbf{e}_b)^\top \mathbf{L}^+ (\mathbf{e}_a - \mathbf{e}_b) \tag{7}$$

where $\mathbf{L}^+$ is the pseudoinverse of the graph Laplacian and $\mathbf{e}_a, \mathbf{e}_b$ are indicator vectors for representative nodes in each patch.

### 5.3 Optimal Corridor Placement

We formulate corridor design as a minimum-cost flow problem on the spatial graph. Given a budget of $B$ cells that can be restored to suitable habitat, we solve:

$$\min_{\mathbf{z} \in \{0,1\}^N} \sum_{(a,b)} R_{ab}(\mathbf{z}) \quad \text{s.t.} \quad \sum_i z_i \le B, \quad z_i = 0 \; \forall\, i \in \bigcup_m P_m \tag{8}$$

where $z_i = 1$ indicates cell $i$ is selected for restoration. We approximate this NP-hard problem using a greedy algorithm with submodular optimization guarantees [Nemhauser et al., 1978].

**Algorithm 1** Greedy Corridor Placement

**Require:** Spatial graph $G$, patches $\{P_m\}$, budget $B$
**Ensure:** Corridor cells $C \subseteq V$

1: $C \leftarrow \emptyset$
2: **for** $b = 1$ to $B$ **do**
3: $\quad v^* \leftarrow \arg\max_{v \notin C \cup \bigcup P_m} \Delta R(v \mid C)$
4: $\quad C \leftarrow C \cup \{v^*\}$
5: **end for**
6: **return** $C$

Here $\Delta R(v \mid C)$ is the marginal reduction in total effective resistance from adding cell $v$ to the current corridor set $C$.

# 6 Data and Experimental Setup

## 6.1 Species Occurrence Data

We compiled occurrence records from multiple sources:

- **GBIF** [GBIF, 2024]: 12.4 million georeferenced records for 847 terrestrial vertebrate species.
- **eBird** [Sullivan et al., 2009]: 8.2 million structured bird survey records.
- **Camera trap networks**: 1.3 million detection records from TEAM [Jansen et al., 2014] and Wildlife Insights [Ahumada et al., 2020].

After spatial thinning (minimum 1 km separation) and temporal filtering (2000–2023), the final dataset contains 4.7 million occurrence records.

## 6.2 Environmental Variables

Bioclimatic variables were sourced from WorldClim v2.1 [Fick & Hijmans, 2017] at 30 arc-second resolution. Topographic variables were derived from SRTM v4.1 [Jarvis et al., 2008]. Land cover was from ESA WorldCover 2021 [Zanaga et al., 2022]. MODIS vegetation indices were averaged over 2018–2022.

## 6.3 Climate Projections

We used downscaled CMIP6 projections from WorldClim [Fick & Hijmans, 2017] for four SSP scenarios at three time periods (2041–2060, 2061–2080, 2081–2100). Twelve GCMs were selected based on performance evaluation by Arias et al. [2021].

## 6.4 Study Regions

We evaluate across six biogeographic realms:

1. Neotropical (Amazon basin focus)
2. Afrotropical (East Africa highlands)
3. Indo-Malayan (Southeast Asian tropics)
4. Palearctic (European temperate forests)
5. Nearctic (North American boreal-temperate transition)
6. Australasian (Eastern Australia)

## 6.5 Baseline Models

- **MaxEnt** [Phillips et al., 2006]: Standard configuration with linear, quadratic, hinge, and threshold features.
- **Random Forest (RF)**: 500 trees, following Cutler et al. [2007].
- **BRT**: Boosted regression trees [Elith et al., 2008] with 5000 trees and interaction depth 5.
- **DeepSDM**: 4-layer MLP with 256 hidden units per layer [Botella et al., 2018].
- **CNN-SDM**: Convolutional model on environmental raster patches [Deneu et al., 2021].

### 6.6 Evaluation Protocol

We employ two validation strategies:

1. **Spatial block cross-validation**: Study area divided into $10 \times 10$ blocks, with blocks assigned to 5 folds ensuring minimum 200 km separation between train and test [Roberts et al., 2017].

2. **Temporal validation**: Models trained on 2000–2015 data, validated on 2016–2023 independent resurvey data from standardized monitoring programs.

Metrics include AUC, True Skill Statistic (TSS), and Boyce index for continuous suitability evaluation [Hirzel et al., 2006].

## 7 Results

### 7.1 Current Distribution Accuracy

Table 1: Model performance on current species distributions (mean ± SD across 847 species). Best results in **bold**.

| Model | AUC | TSS | Boyce |
|---|---|---|---|
| MaxEnt | 0.867 ± 0.071 | 0.612 ± 0.114 | 0.751 ± 0.098 |
| Random Forest | 0.881 ± 0.063 | 0.641 ± 0.107 | 0.773 ± 0.091 |
| BRT | 0.889 ± 0.058 | 0.658 ± 0.102 | 0.789 ± 0.087 |
| DeepSDM | 0.901 ± 0.054 | 0.687 ± 0.096 | 0.812 ± 0.079 |
| CNN-SDM | 0.912 ± 0.049 | 0.711 ± 0.091 | 0.834 ± 0.072 |
| **HabitatNet** | **0.943 ± 0.038** | **0.762 ± 0.078** | **0.891 ± 0.056** |

HabitatNet achieves a mean AUC of 0.943, representing a 3.4% absolute improvement over the next-best model (CNN-SDM). The improvement is most pronounced for species with fragmented ranges, where graph-based spatial reasoning provides the greatest advantage.

### 7.2 Temporal Transferability

Table 2: Temporal validation: models trained on 2000–2015, evaluated on 2016–2023 resurvey data.

| Model | AUC | TSS | Δ AUC |
|---|---|---|---|
| MaxEnt | 0.793 ± 0.089 | 0.497 ± 0.131 | −0.074 |
| Random Forest | 0.801 ± 0.082 | 0.513 ± 0.124 | −0.080 |
| BRT | 0.811 ± 0.077 | 0.531 ± 0.119 | −0.078 |
| DeepSDM | 0.834 ± 0.069 | 0.567 ± 0.108 | −0.067 |
| CNN-SDM | 0.849 ± 0.063 | 0.591 ± 0.101 | −0.063 |
| **HabitatNet** | **0.907 ± 0.047** | **0.689 ± 0.084** | **−0.036** |

HabitatNet exhibits the smallest performance degradation under temporal transfer ($\Delta$AUC $= -0.036$), suggesting superior generalization to novel environmental conditions. The graph structure allows the model to propagate information about distributional shifts through connected landscape elements.

### 7.3 Climate Projection Results

Under SSP2-4.5 (intermediate emissions), HabitatNet projects:

- **Range contractions**: 67% of species lose >10% of current suitable habitat by 2081–2100.
- **Range shifts**: Mean poleward shift of 6.1 km/decade (±2.3), consistent with observed trends [Chen et al., 2011].
- **Elevational shifts**: Mean upslope shift of 11.2 m/decade in mountainous regions.
- **Novel disconnections**: 23% of currently connected habitat networks become fragmented by 2061–2080.

Under SSP5-8.5 (high emissions), projections intensify dramatically: range contractions affect 89% of species, and 41% of habitat networks become disconnected.

### 7.4 Connectivity Analysis

Table 3: Habitat connectivity metrics under current and projected conditions (SSP2-4.5, 2081–2100). Mean across species.

| **Metric** | **Current** | **2081–2100** |
|---|---|---|
| Number of patches | 4.2 ± 2.1 | 7.8 ± 3.6 |
| Largest patch area ($km^2$) | 84,300 ± 41,200 | 52,100 ± 29,800 |
| Effective mesh size | 0.71 ± 0.18 | 0.43 ± 0.21 |
| Graph connectivity index | 0.84 ± 0.11 | 0.58 ± 0.19 |

### 7.5 Corridor Identification

The greedy corridor algorithm (Algorithm 1) identifies restoration priorities that maximize connectivity recovery. With a budget of 5% additional restored area, corridors recover 72% of lost connectivity on average, compared to 31% for random placement and 54% for least-cost path heuristics.

## 8 Case Study: East African Montane Species

We present a detailed case study of 43 montane bird species in the Eastern Arc Mountains and Albertine Rift of East Africa. This region hosts exceptional endemism but faces severe habitat fragmentation from agricultural expansion [Newmark, 2002].

### 8.1 Current Distribution Modeling

HabitatNet identifies 12 distinct montane forest habitat patches, ranging from 23 $km^2$ (Nguru Mountains) to 4,800 $km^2$ (Albertine Rift complex). Species richness peaks at 1,800–2,400 m elevation, with sharp declines above 3,200 m and below 1,200 m.

### 8.2 Climate-Driven Range Shifts

Under SSP2-4.5 by 2081–2100, the model projects:

- Mean upslope shift of the suitable climate envelope by 340 m (±120 m).

- Loss of 38% of currently suitable montane forest area due to "summit trap" effects—species near mountain peaks cannot shift further upward [Freeman et al., 2018].
- Complete loss of suitable habitat for 4 narrow-endemic species with elevation ranges <500 m.

### 8.3 Priority Corridors

The corridor algorithm identifies three critical connectivity zones:

1. **Uluguru–Udzungwa link**: A 47 km lowland gap currently acting as a dispersal barrier, where targeted restoration of 12 $km^2$ of riparian forest would connect 230 $km^2$ of montane habitat.
2. **Albertine Rift mid-elevation belt**: A 2,100 m elevation corridor linking three national parks.
3. **Usambara–Pare bridge**: Restoration of 8 $km^2$ of degraded forest to reconnect two endemic bird areas.

## 9 Discussion

### 9.1 Advantages of Graph-Based Modeling

HabitatNet's spatial graph representation provides three key advantages over traditional SDMs:

**Spatial context.** Message passing aggregates information from neighboring cells, allowing the model to recognize habitat edges, ecotones, and the spatial configuration of suitable patches—information invisible to point-based models.

**Learned connectivity.** By learning edge weights from occurrence data, HabitatNet discovers dispersal barriers and corridors that may not be apparent from land cover alone. Analysis of learned edge weights reveals that the model assigns high resistance to agricultural land (mean conductance 0.12), moderate resistance to secondary forest (0.47), and low resistance to primary forest (0.91).

**Scale-aware predictions.** Multi-hop message passing allows the model to incorporate information at multiple spatial scales without explicit multi-scale feature engineering.

### 9.2 Limitations

Several limitations warrant consideration:

1. **Biotic interactions**: HabitatNet models abiotic habitat suitability and does not explicitly account for species interactions such as competition, predation, or mutualism.
2. **Adaptive capacity**: The model does not represent genetic adaptation or phenotypic plasticity, potentially overestimating range loss for adaptable species.
3. **Land-use change**: Our base projections use static land cover. Integration with land-use change scenarios (LUH2) is straightforward but introduces additional uncertainty.
4. **Computational cost**: Graph construction and GNN training on continental-scale grids require significant computational resources (approximately 48 GPU-hours for a full training run on the Neotropical realm).

5. **Data bias**: GBIF occurrence records are spatially biased toward accessible areas and wealthy nations [Beck et al., 2014]. While spatial thinning and bias correction mitigate this, residual effects may persist.

### 9.3 Conservation Implications

Our results have direct implications for conservation planning:

1. **Protected area expansion**: Current protected area networks cover only 34% of projected future habitat, suggesting significant gaps in anticipatory conservation planning.
2. **Corridor investment**: Targeted corridor restoration offers high returns—recovering 72% of lost connectivity with only 5% additional area—supporting investment in landscape-scale conservation.
3. **Climate refugia**: HabitatNet identifies climate refugia—areas of persistent suitability across scenarios—that merit highest protection priority.

### 9.4 Open Science and Reproducibility

All code, trained models, and processed datasets are released through the Zoo Open Science Network under Apache 2.0 and CC-BY-4.0 licenses, respectively. The framework is available as a Python package (`habitatnet`) with documentation and tutorials. Model training is fully reproducible through provided configuration files and random seeds.

## 10 Implementation Details

### 10.1 Software Stack

HabitatNet is implemented in Python 3.11 using PyTorch 2.2 and PyTorch Geometric 2.5. Spatial data processing uses `rasterio`, `geopandas`, and `h3-py` for hexagonal grid construction. The corridor algorithm uses `networkx` for graph operations and `scipy.sparse` for efficient Laplacian computations.

### 10.2 Training Protocol

- **Optimizer**: AdamW with learning rate $3 \times 10^{-4}$, weight decay $10^{-4}$.
- **Schedule**: Cosine annealing with warm restarts over 200 epochs.
- **Batch size**: 16 subgraphs of 10,000 nodes each, sampled using ClusterGCN [Chiang et al., 2019].
- **Graph layers**: $L = 6$ message-passing layers with 128 hidden dimensions.
- **Regularization**: Dropout (0.2), edge dropout (0.1), and DropMessage [Fang et al., 2023].
- **Loss weights**: $\lambda_1 = 0.1$, $\lambda_2 = 0.05$ (selected by validation).

### 10.3 Computational Requirements

Training a continental-scale model requires approximately 48 GPU-hours on an NVIDIA A100 (80 GB). Inference for a 1-million-cell region takes approximately 4 minutes. The corridor algorithm runs in $O(B \cdot N \cdot M^2)$ time, where $B$ is the budget, $N$ is the number of cells, and $M$ is the number of patches.

## 11 Future Work

Several directions extend this work:

1. **Dynamic land use**: Coupling HabitatNet with land-use change models (e.g., CLUE, LUH2) for joint climate-land use projections.
2. **Species interactions**: Incorporating biotic interaction networks as an additional graph layer, enabling prediction of community-level responses to climate change.
3. **Uncertainty quantification**: Developing Bayesian GNN variants for principled uncertainty estimation across model, data, and scenario uncertainty sources.
4. **Real-time monitoring**: Integrating satellite-derived near-real-time land cover change detection to update habitat maps continuously.
5. **Decentralized compute**: Deploying training workloads on the Zoo Open Science Network's decentralized compute infrastructure, enabling community-contributed computational resources.
6. **DAO-governed priorities**: Allowing the Zoo DAO to vote on conservation priority regions and species through quadratic voting mechanisms (see ZIP-003).

## 12 Conclusion

We have presented HabitatNet, a graph neural network framework for species distribution modeling that jointly predicts habitat suitability, landscape connectivity, and optimal migration corridors under climate change. Evaluated on 847 species across six biogeographic realms, HabitatNet substantially outperforms existing approaches in both current prediction accuracy and temporal transferability. The framework provides actionable conservation guidance by identifying connectivity bottlenecks and prioritizing corridor restoration investments.

As climate change accelerates the redistribution of life on Earth, tools that bridge machine learning and conservation biology become increasingly critical. HabitatNet represents a step toward computationally grounded, spatially explicit conservation planning at the scales required by the biodiversity crisis. Through Zoo Labs Foundation's commitment to open science and decentralized research infrastructure, we ensure that these tools remain accessible to the global conservation community.

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# A Supplementary: Feature Importance Analysis

We analyze feature importance using integrated gradients [Chen et al., 2020] applied to the feature encoder. Across all 847 species, the most influential variables are:

Table 4: Top 10 features by mean integrated gradient magnitude across all species.

| **Rank** | **Feature** | **Importance** |
|---|---|---|
| 1 | Annual mean temperature (BIO1) | 0.187 |
| 2 | Annual precipitation (BIO12) | 0.156 |
| 3 | Elevation | 0.134 |
| 4 | Tree cover fraction | 0.121 |
| 5 | Temperature seasonality (BIO4) | 0.098 |
| 6 | NDVI mean | 0.089 |
| 7 | Precipitation seasonality (BIO15) | 0.076 |
| 8 | Distance to water | 0.062 |
| 9 | Slope | 0.048 |
| 10 | Temperature annual range (BIO7) | 0.041 |

Feature importance varies substantially across taxa: for amphibians, precipitation variables dominate; for birds, temperature and NDVI are most influential; for mammals, tree cover and elevation are primary drivers.

# B Supplementary: Hyperparameter Sensitivity

Table 5: Sensitivity of HabitatNet AUC to key hyperparameters.

| **Parameter** | **Values tested** | **Best** | **AUC range** |
|---|---|---|---|
| Message-passing layers $L$ | 2, 4, 6, 8, 10 | 6 | 0.931–0.943 |
| Hidden dimension | 64, 128, 256 | 128 | 0.937–0.943 |
| Learning rate | $10^{-3}$, $3 \times 10^{-4}$, $10^{-4}$ | $3 \times 10^{-4}$ | 0.935–0.943 |
| Spatial loss $\lambda_1$ | 0.01, 0.05, 0.1, 0.5 | 0.1 | 0.938–0.943 |
| Connectivity loss $\lambda_2$ | 0.01, 0.05, 0.1 | 0.05 | 0.940–0.943 |

The model is most sensitive to the number of message-passing layers and least sensitive to the connectivity loss weight, suggesting that the spatial structure provides the dominant signal.